HW1: Regression Modeling

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due on 10/11 (Tue)

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Before working on HW1, you better complete the practice of ISLR ch2-3 R Labs. In particular, you may follow similar fitting steps demonstrated in ISLR ch3 R Lab for the following two data analysis.

1 Problem1

The first data consist of 4 variables (daily measurements) measured at the same place:

- ozone: response variable
- radiation
- temperature
- wind (speed)

Goal: Predict ozone using regression models.

Your analysis should include the following:

(a) Exploratory data analysis (EDA) among 4 variables
(b) Regression model fitting and model summaries.
(c) Model selection and diagonostics
(d) Comments on your prediction results and scientific findings. (state at least 3 viewpoints with data evidence)

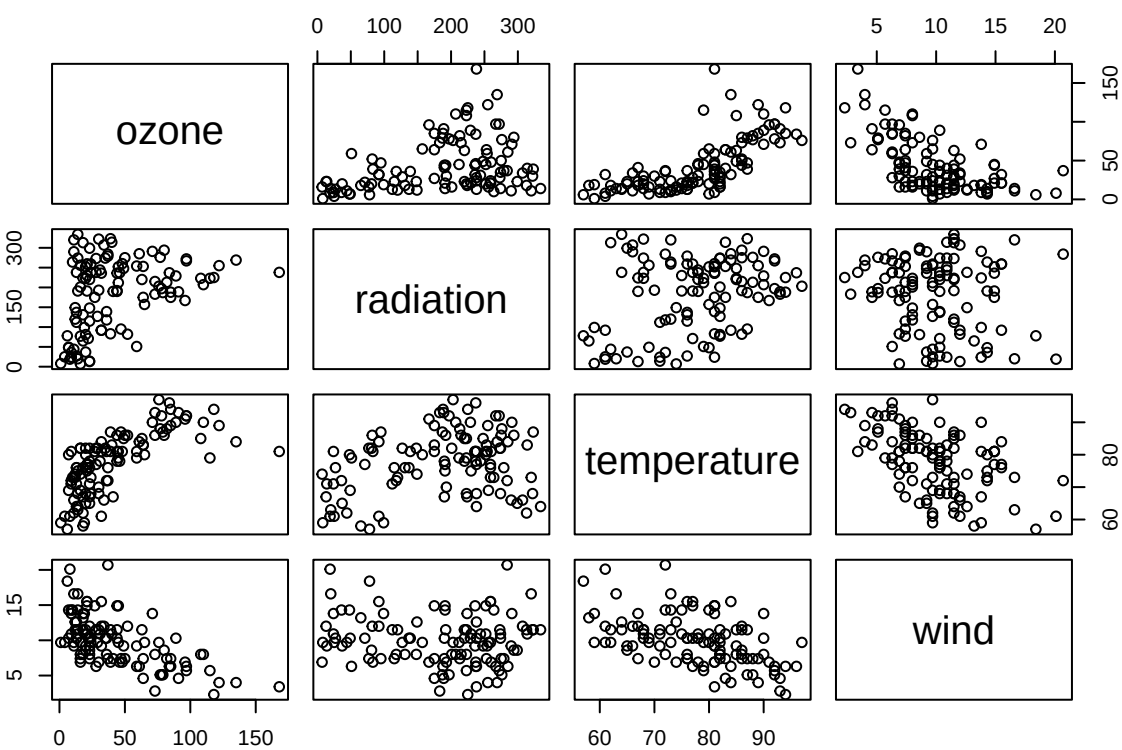
```
dat1 <- read.csv("ozone.csv")
head(dat1)
```

```
##   ozone radiation temperature wind
## 1    41       190          67  7.4
## 2    36       118          72  8.0
## 3    12       149          74 12.6
## 4    18       313          62 11.5
## 5    23       299          65  8.6
## 6    19        99          59 13.8
```

```
dim(dat1)
```

```
## [1] 111   4
```

```
pairs(dat1)
```



View data and start your analysis here…

1.1 Exploratory Data Analysis

Data Description

This is a dataset consisting of 111 observations and four continuous variables. There is no any missing value present. Table 1 shows the several basic statistics, including minimum, quantiles, maximum, mean and median, for each of the variables.

```
stat = round(sapply(dat1, summary),2)
kable(
  as.data.frame(stat),
  format = "latex",
  booktabs = TRUE,
  align = "c",
  col.names = c("ozone", "radiation", "temperature", "wind"),
  caption = "Summary for the variables",
  escape = FALSE,
  linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 1: Summary for the variables

	ozone	radiation	temperature	wind
Min.	1.0	7.0	57.00	2.30
1st Qu.	18.0	113.5	71.00	7.40
Median	31.0	207.0	79.00	9.70
Mean	42.1	184.8	77.79	9.94
3rd Qu.	62.0	255.5	84.50	11.50
Max.	168.0	334.0	97.00	20.70

From Table 1, one can see that the distributions of temperature and wind seem to be symmetric since their means do not differ a lot form their medians in consideration of their large ranges. In contrast, radiation seems to be somewhat skewed to the left, and ozone seems to be somewhat skewed to the right since their means are either much smaller or larger than their medians in consideration of their small ranges. Moreover, the skewness seems to be especially extreme for ozone.

Data Feature

To explore the basic relationships among the seven continuous variables, variable visualization (by scatter plots) and correlation analysis are first conducted as follows.

```
pairs.panels(dat1[c(2,3,4,1)],
             smooth = TRUE,
             scale = FALSE,
             density = TRUE,
             ellipses = TRUE,
             method = "kendall",
             pch = 21,
             lm = FALSE,
             cor = TRUE,
             jiggle = FALSE,
             factor = 2,
```

```
            hist.col = 4,
            stars = TRUE,
            ci = TRUE,
            cex=0.8)
```

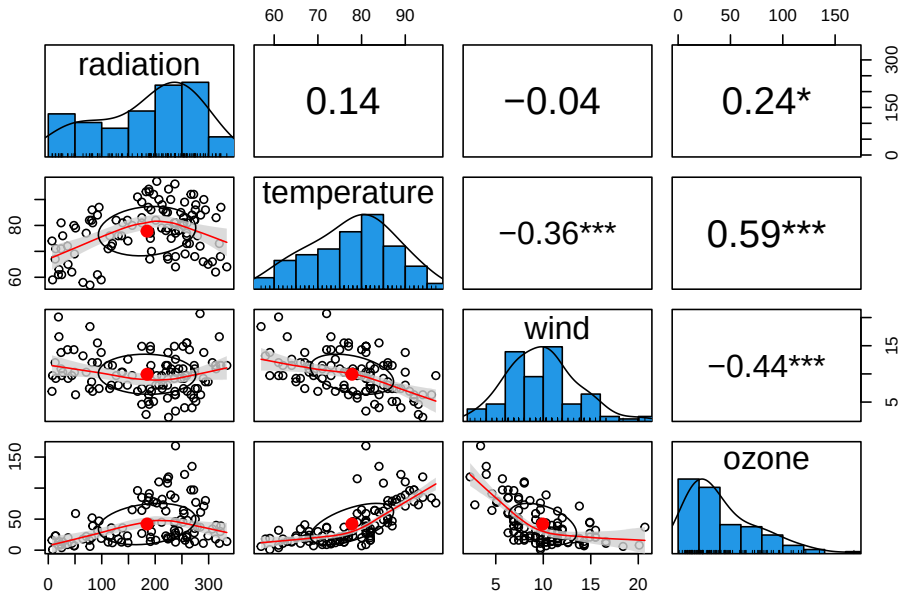


Figure 1: Correlation plot for the four variables

As shown in the scatter plots in Figure 1, since no any pair of the variables seems to follow bivariate normal distribution, Kendall's τ statistic is used to measure the association among the variables. In the upper triangle of Figure 1, the values of the Kendall's τ statistics are shown. One can see that radiation, temperature and wind are all significantly correlated with ozone according to Kendall's test with significance level $\alpha = 0.05$. More specifically, the association between ozone and radiation and temperature are positive and that between ozone and wind is negative. Moreover, it is also worth noticing that temperature and wind are significantly correlated according to Kendall's test, where the association is negative.

In each scatter plot in Figure 1, there is a red line showing a LOESS smoothed fitted line. When concentrating on the relationships between ozone and radiation, temperature and wind (the last row in Figure 1), one can see that the red lines are all curved, which may suggest curvature relationships between ozone and the remaining variables. Hence, when the interest is to predict ozone (response) by the remaining variables (predictors), it is reasonable to start with a linear model with quadratic term, that is

$$ozone = \beta_0 + \beta_1\, radiation + \beta_2\, temperature + \beta_3\, wind + \beta_{11}\, radiation^2 + \beta_{22}\, temperature^2 + \beta_{33}\, wind^2 + \epsilon. \quad (1)$$

In addition, from the scatter plots and the histograms, one may suspected that there may be respectively two populations in the data with respect to radiation and wind. In the other words, two indicator variables may be needed to discriminate out the four populations, which can thus leave more flexibility to the model. To construct such indicator variables for radiation and wind, a clustering analysis based on PAM algorithm is conducted. Figure 2 shows the histograms of radiation and wind after clustering.

```
pam.res = pam(dat1[,c(2)], 2)
dat1 = cbind(dat1, radiation2 = pam.res$clustering)
pam.res = pam(dat1[,c(4)], 2)
dat1 = cbind(dat1, wind2 = pam.res$clustering)
```

```
dat1$radiation2 = dat1$radiation2+1
dat1$radiation2 = ifelse(dat1$radiation2 == 3, 1, 2)
```

```
par(mfrow = c(1,4))
c1 = rgb(173,216,230,max = 255, alpha = 80, names = "lt.blue")
c2 = rgb(255,192,203, max = 255, alpha = 80, names = "lt.pink")
hist(dat1[dat1$radiation2 == 1,]$radiation, freq = FALSE, col = c1, main = "cluster 1", xlab = "radiati
lines(density(dat1[dat1$radiation2 == 1,]$radiation), lwd = 2)
hist(dat1[dat1$radiation2 == 2,]$radiation, freq = FALSE, col = c2, main = "cluster 2", xlab = "radiati
lines(density(dat1[dat1$radiation2 == 2,]$radiation), lwd = 2)
hist(dat1[dat1$wind2 == 1,]$wind, freq = FALSE, col = c1, main = "cluster 1", xlab = "wind")
lines(density(dat1[dat1$wind2 == 1,]$wind), lwd = 2)
hist(dat1[dat1$wind2 == 2,]$wind, freq = FALSE, col = c2, main = "cluster 2", xlab = "wind")
lines(density(dat1[dat1$wind2 == 2,]$wind), lwd = 2)
```

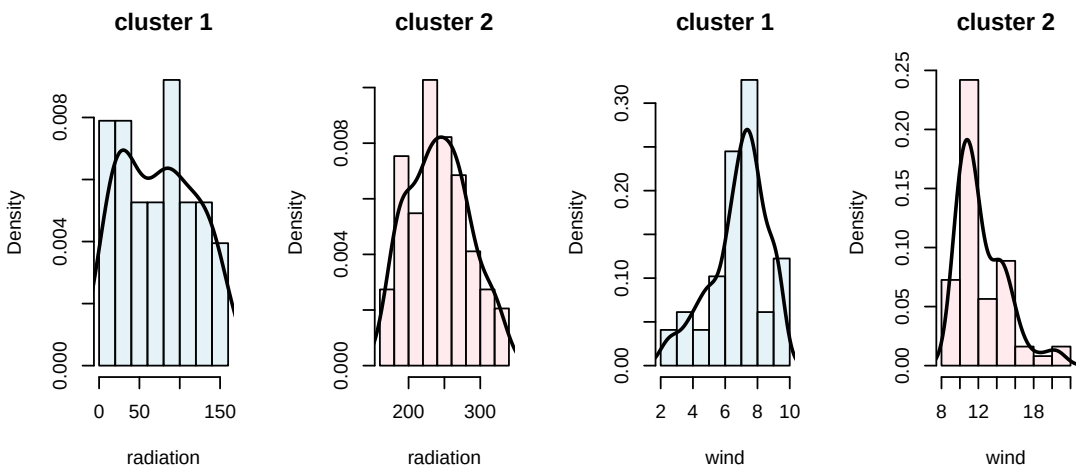


Figure 2: Histograms of radiation and wind after clustering

From Figure 2, one can see that the distributions for radiation and wind have only one mode respectively except for cluster 1 of radiation. Overall, this clustering result implies that the data can be reasonably regarded as consisting of 4 populations, each of which is characterized by a certain combination of the clusters on radiation and wind. Furthermore, one can see that for both radiation and wind, the values in the cluster 1 are obviously smaller than those in the cluster 2. Hence, it is reasonable to name the cluster 1 as "low level" and the cluster 2 as "high level". Two indicator variables named "radiation2" and "wind2" are added in the dataset to classify the two clusters (levels) in radiation and wind, which are defined as 0 for cluster 1 and 1 for cluster 2.

```
par(mfrow =c(1,2), mar = c(5, 4, 4, 1) + 0.1)
plot(dat1$radiation, dat1$ozone, xlab = "radiation", ylab = "ozone")
points(dat1[dat1$radiation2==1,]$radiation, dat1[dat1$radiation2==1,]$ozone, pch= 19, col = "deepskyblu
```

```
points(dat1[dat1$radiation2==2,]$radiation, dat1[dat1$radiation2==2,]$ozone, pch= 19, col = "indianred1
legend("topright", c("low", "high"), pch = 19, col = c("deepskyblue3", "indianred1"))

plot(dat1$wind, dat1$ozone, xlab = "wind", ylab = "ozone")
points(dat1[dat1$wind2==1,]$wind, dat1[dat1$wind2==1,]$ozone, pch= 19, col = "deepskyblue3")
points(dat1[dat1$wind2==2,]$wind, dat1[dat1$wind2==2,]$ozone, pch= 19, col = "indianred1")
legend("topright", c("low", "high"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

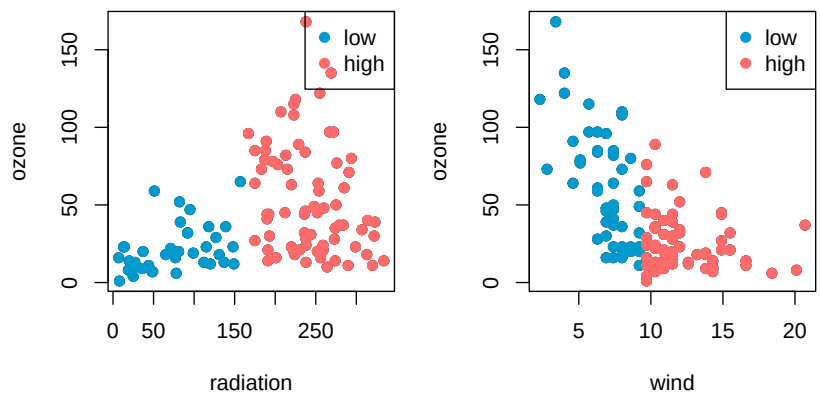


Figure 3: Conditional relationships between ozone and radiation and wind

Figure 3 shows the scatter plots in which the clusters in radiation and wind are discriminated by different colors. The blue points denote for the observations with low radiation or low wind (speed), and the red points denote for those with high radiation or high wind (speed). The vertical-axis refers to ozone in each plot. One can see that the association between radiation and ozone seems to be different conditioned on each level of radiation. Similarly, the association between wind and ozone also seems to depend on the levels of wind. This fact implies that it is worthy to investigate a linear model in which radiation2 and wind2 are also included, that is

$$\begin{aligned} ozone = \beta_0 + \beta_1\, radiation + \beta_2\, temperature + \beta_3\, wind \\ + \beta_4\, radiation2 + \beta_5\, wind2 + \beta_{14}\, radiation2 \times radiation + \beta_{35}\, wind2 \times wind + \epsilon. \end{aligned} \tag{2}$$

1.2 Regression model fitting, model summaries and model diagnostics

Following the discussion above, Model (1) and Model (2) are fitted. Based on the fitted results, several remedial measures may be taken to revise the models. During the process, interpretability is regarded as the most important, and the prediction performance is considered in the second place. Three assumptions are made for the error term ϵ in the models: homogeneity, normality and independence. Furthermore, the following analyses which involve hypothesis testing are all conducted under significance level $\alpha = 0.05$.

1.2.1 Model (1)

Summary for fitted Model (1)

Table 2 shows the summary for fitted Model (1), from which one can see that the quadratic term of radiation is not significant. Moreover, the signs of the estimates for each coefficients are all consistent with those observed in Figure 1, implying that no serious problems such as multicollinearity occurred during the fitting process.

```
fit1 = lm(ozone~poly(radiation,2)+poly(temperature,2)+poly(wind,2), data = dat1)
```

```
summ = as.data.frame(round(summary(fit1)$coefficients,4))
rownames(summ) = c("Intercept","radiation","$\\text{radiation}^2$",
                   "temperature","$\\text{temperature}^2$","wind","$\\text{wind}^2$")
kable(summ,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Estimate","SE","t-value","p-value"),
  align = c("c", "c", "c", "c"),
  escape = FALSE,
  linesep = "",
  caption = "Summary for model 1") %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 2: Summary for model 1

	Estimate	SE	t-value	p-value
Intercept	42.0991	1.7347	24.2690	0.0000
radiation	65.0851	19.3787	3.3586	0.0011
radiation^2	-16.2587	20.1743	-0.8059	0.4221
temperature	142.7080	23.5024	6.0721	0.0000
temperature^2	56.0429	19.1384	2.9283	0.0042
wind	-125.8000	21.3908	-5.8810	0.0000
wind^2	88.6357	19.1991	4.6166	0.0000

Diagnostics for Model (1)

However, from Figure 4 one can see that neither the homogeneity nor the normality assumption do seem to be acceptable for the error term, implying that the p-values may not be meaningful. In fact, the Breusch-Pagan test, Shapiro-Wilk test and Durbin-Watson test are conducted to examine the homogeneity, normalily and independence respectively (see 3), which show consistent results with Figure 4. Moreover, Figure 5 shows the patterns of residuals with respect to each of the variables, implying that some transformation on the response ozone may be helpful.

```
par(mfrow = c(2,2), mar =c(4,4,3,1)+0.1)
plot(fit1)
```

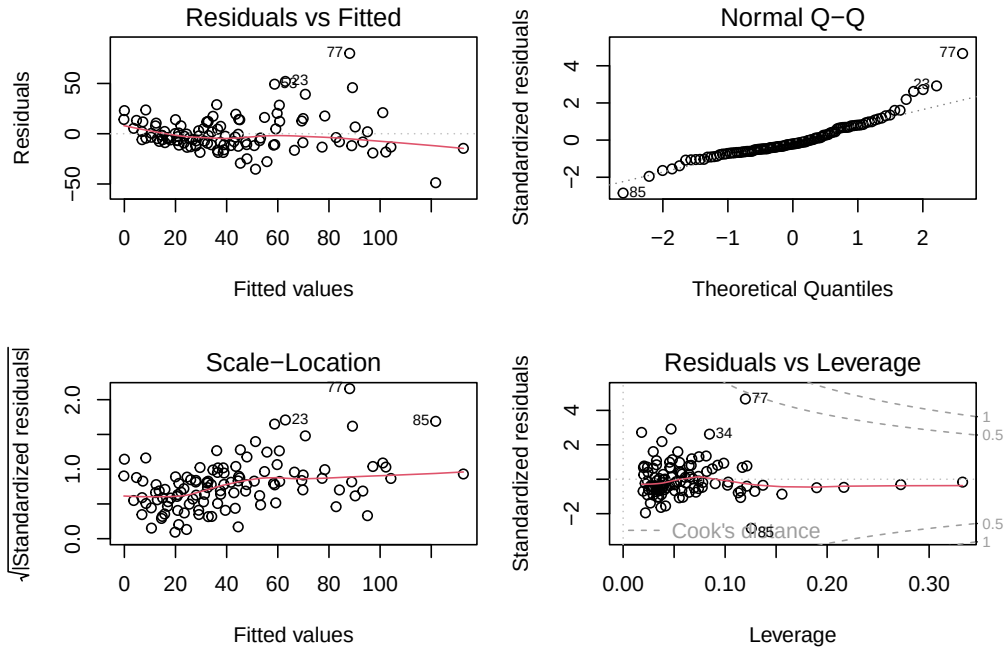


Figure 4: Residual analysis

```
resids = fit1$residuals
bp = bptest(fit1)
sh = shapiro.test(resids)
db = durbinWatsonTest(fit1)

rtest = t(round(as.data.frame(c(bp$statistic, bp$p.value, sh$statistic, sh$p.value, db$dw, db$p)),3))
rownames(rtest) = NULL

kable(
  rtest,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = rep(c("test statistic", "p-value"),3),
  align = "c",
  caption = "Model diagnostics",
  escape = FALSE
) %>%
add_header_above(c("Breusch-Pagan" = 2, "Shapiro-Wilk" = 2, "Durbin-Watson" = 2))
```

Table 3: Model diagnostics

Breusch-Pagan		Shapiro-Wilk		Durbin-Watson	
test statistic	p-value	test statistic	p-value	test statistic	p-value
25.371	0	0.911	0	1.777	0.184

```
par(mfrow = c(1,3))
for(i in 2:4){
 plot(dat1[,i], fit1$residuals, xlab = names(dat1)[i], ylab = "residuals")
}
points(dat1[dat1$wind2==1,]$wind, fit1$residuals[dat1$wind2==1], pch= 19, col = "deepskyblue3")
points(dat1[dat1$wind2==2,]$wind, fit1$residuals[dat1$wind2==2], pch= 19, col = "indianred1")
legend("topright", c("low", "high"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

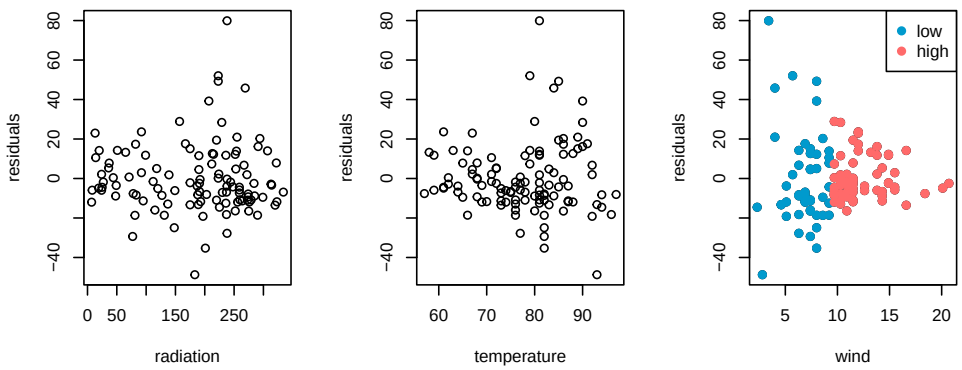


Figure 5: Residual analysis

Remedial measure for Model (1)

```
dat1$radiation2 = factor(dat1$radiation2, levels = c(1,2))
dat1$wind2 = factor(dat1$wind2, levels = c(1,2))
```

Box-Cox transformation which suggested power of 0.2 is used to transform the response ozone. After the diagnostics, it was found that all the assumptions become acceptable. Hence, it was meaningful to comment on the p-values. The quadratic term of radiation and temperature were not significant, and temperature^2 was the least significant one. Hence temperature^2 was omitted from the model. Finally the quadratic term of radiation is still not significant, but it is reasonable to let it stay in the model. The reason is not only because that its p-value is very closed to 0.05 but also due to the fact that the homogeneity assumption would be no longer acceptable once radiation^2 were omitted. The summary for the revised version of fitted Model (1) is shown in Table 4, and the diagnostic results are shown in Figure 6, Figure 7 and Table 5.

```
fit1 = lm(ozone^0.2~poly(radiation,2)+temperature+poly(wind,2), data = dat1)
```

```
summ = as.data.frame(round(summary(fit1)$coefficients,4))
rownames(summ) = c("Intercept","radiation","$\\text{radiation}^2$",
                   "temperature", "wind", "$\\text{wind}^2$")
kable(summ,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Estimate","SE","t-value","p-value"),
  align = c("c", "c", "c", "c"),
  escape = FALSE,
  linesep = "",
```

```
  caption = "Summary for model 1 of revised version") %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 4: Summary for model 1 of revised version

	Estimate	SE	t-value	p-value
Intercept	0.7353	0.1816	4.0480	0.0001
radiation	0.9003	0.1908	4.7184	0.0000
radiation^2	-0.3563	0.1989	-1.7913	0.0761
temperature	0.0164	0.0023	7.0433	0.0000
wind	-1.0400	0.2100	-4.9533	0.0000
wind^2	0.6610	0.1835	3.6013	0.0005

```
par(mfrow = c(2,2), mar =c(4,4,3,1)+0.1)
plot(fit1)
```

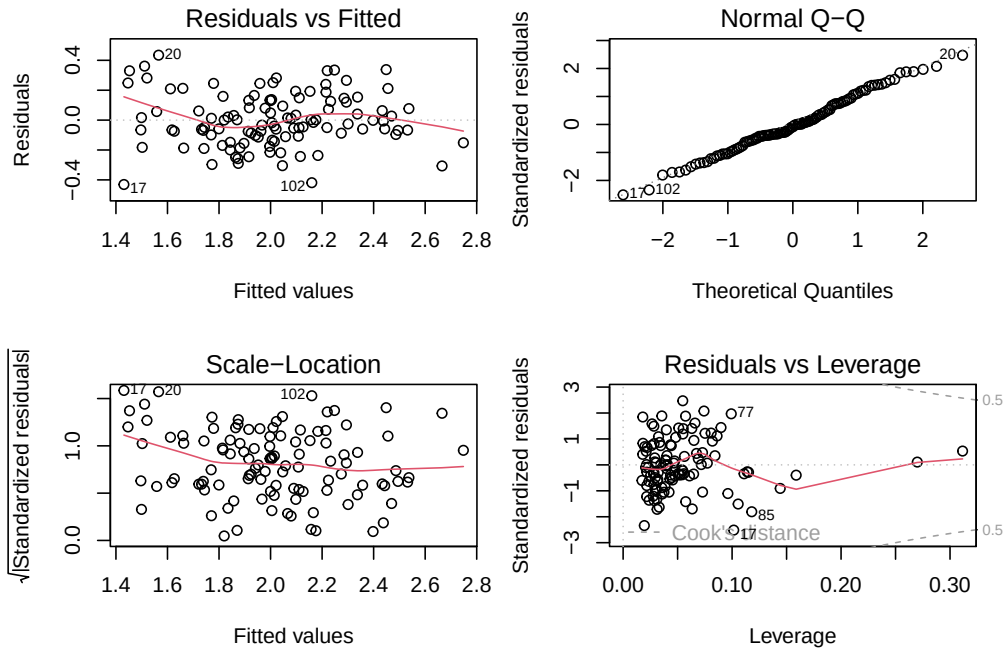


Figure 6: Residual analysis

```
resids = fit1$residuals
bp = bptest(fit1)
sh = shapiro.test(resids)
db = durbinWatsonTest(fit1)

rtest = t(round(as.data.frame(c(bp$statistic, bp$p.value, sh$statistic, sh$p.value, db$dw, db$p)),3))
rownames(rtest) = NULL
```

```r
kable(
  rtest,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = rep(c("test statistic", "p-value"),3),
  align = "c",
  caption = "Model diagnostics",
  escape = FALSE
) %>%
add_header_above(c("Breusch-Pagan" = 2, "Shapiro-Wilk" = 2, "Durbin-Watson" = 2))
```

Table 5: Model diagnostics

Breusch-Pagan		Shapiro-Wilk		Durbin-Watson	
test statistic	p-value	test statistic	p-value	test statistic	p-value
10.614	0.06	0.99	0.589	1.744	0.136

```r
par(mfrow = c(1,3))
for(i in 2:4){
 plot(dat1[,i], fit1$residuals, xlab = names(dat1)[i], ylab = "residuals")
}
points(dat1[dat1$wind2==1,]$wind, fit1$residuals[dat1$wind2==1], pch= 19, col = "deepskyblue3")
points(dat1[dat1$wind2==2,]$wind, fit1$residuals[dat1$wind2==2], pch= 19, col = "indianred1")
legend("topright", c("low", "high"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

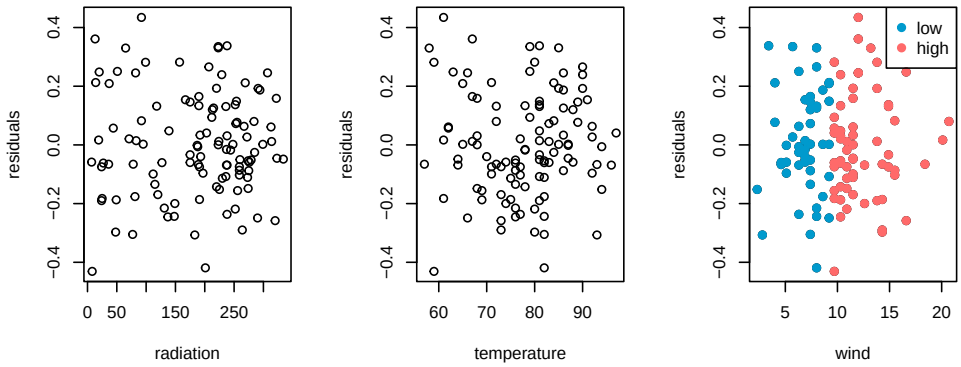


Figure 7: Residual analysis (cont.)

1.2.2 Model (2)

Summary for fitted Model (2)

Table 6 shows the summary for fitted Model (2), from which one can see that all of the terms involved in radiation are not significant, implying that problems such as multicollinearity may occur during the fitting process.

```
fit2 = lm(ozone~.+radiation2:radiation+wind2:wind, data = dat1)
```

```
summ = as.data.frame(round(summary(fit2)$coefficients,4))
rownames(summ) = c("Intercept","radiation","temperature", "wind", "radiation2", "wind2", "radiation2$\\
kable(summ,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Estimate","SE","t-value","p-value"),
  align = c("c", "c", "c", "c"),
  escape = FALSE,
  linesep = "",
  caption = "Summary for model 2") %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 6: Summary for model 2

	Estimate	SE	t-value	p-value
Intercept	7.2318	25.0255	0.2890	0.7732
radiation	0.0109	0.0707	0.1538	0.8781
temperature	1.3504	0.2482	5.4401	0.0000
wind	-9.7111	1.7041	-5.6985	0.0000
radiation2	9.3184	16.2216	0.5744	0.5669
wind2	-83.7474	17.1569	-4.8813	0.0000
radiation2×radiation	0.0054	0.0935	0.0581	0.9538
wind2×wind	9.2647	1.9259	4.8105	0.0000

Diagnostics for Model (2)

Moreover, from Figure 8 one can see that neither the homogeneity nor the normality assumption do seem to be acceptable for the error term, implying that the p-values may not be meaningful. The Breusch-Pagan test, Shapiro-Wilk test and Durbin-Watson test are also conducted to examine the assumptions (see 7), which show consistent results with Figure 8. Similar to before, 9 shows the patterns of residuals with respect to each of the variables which implies that some transformation on the response ozone may be helpful.

```
par(mfrow = c(2,2), mar =c(4,4,3,1)+0.1)
plot(fit2)
```

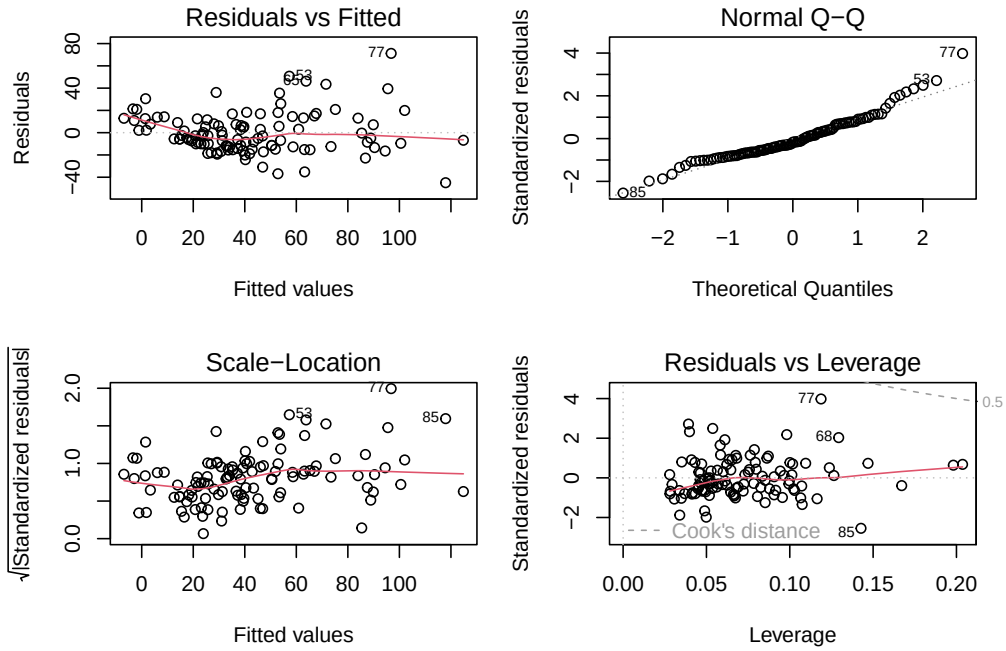


Figure 8: Residual analysis

```
resids = fit2$residuals
bp = bptest(fit2)
sh = shapiro.test(resids)
db = durbinWatsonTest(fit2)

rtest = t(round(as.data.frame(c(bp$statistic, bp$p.value, sh$statistic, sh$p.value, db$dw, db$p)),3))
rownames(rtest) = NULL

kable(
  rtest,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = rep(c("test statistic", "p-value"),3),
  align = "c",
  caption = "Model diagnostics",
  escape = FALSE
) %>%
add_header_above(c("Breusch-Pagan" = 2, "Shapiro-Wilk" = 2, "Durbin-Watson" = 2))
```

Table 7: Model diagnostics

Breusch-Pagan		Shapiro-Wilk		Durbin-Watson	
test statistic	p-value	test statistic	p-value	test statistic	p-value
18.65	0.009	0.949	0	1.68	0.064

```
par(mfrow = c(1,3))
for(i in 2:4){
 plot(dat1[,i], fit2$residuals, xlab = names(dat1)[i], ylab = "residuals")
}
points(dat1[dat1$wind2==1,]$wind, fit2$residuals[dat1$wind2==1], pch= 19, col = "deepskyblue3")
points(dat1[dat1$wind2==2,]$wind, fit2$residuals[dat1$wind2==2], pch= 19, col = "indianred1")
legend("topright", c("low", "high"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

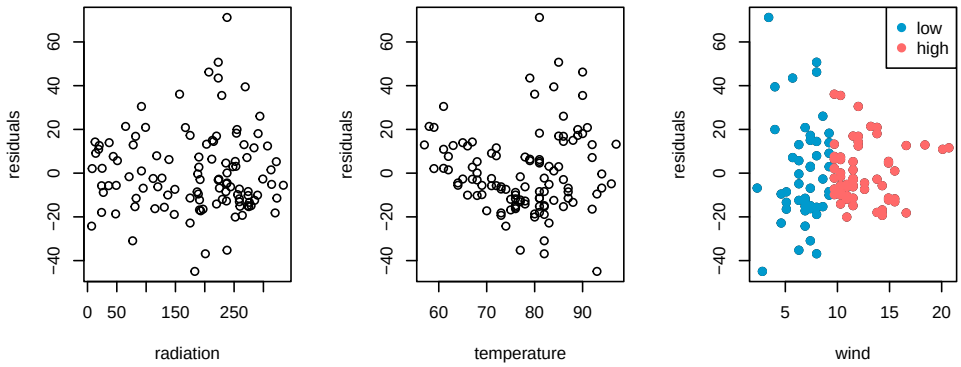


Figure 9: Residual analysis

Remedial measure for Model (2)

Box-Cox transformation which suggested power of 0.2 is used to transform the response ozone. After the diagnostics, it was found that all the assumptions become acceptable. Hence, it was meaningful to comment on the p-values. The terms involved radiation were still not significant. Hence radiation2×radiation and radiation2 were sequentially omitted from the model. The summary for the revised version of fitted Model (2) is shown in Table 8, and the diagnostic results are shown in Figure 10, Figure 11 and Table 9.

```
fit2 = update(fit2, ozone^0.2~.-(radiation2:radiation)-radiation)
```

```
summ = as.data.frame(round(summary(fit2)$coefficients,4))
rownames(summ) = c("Intercept","temperature", "wind", "radiation2", "wind2", "wind2$\\times$wind")
kable(summ,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Estimate","SE","t-value","p-value"),
  align = c("c", "c", "c", "c"),
  escape = FALSE,
  linesep = "",
  caption = "Summary for model 2 of revised version") %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

```
par(mfrow = c(2,2), mar =c(4,4,3,1)+0.1)
plot(fit2)
```

Table 8: Summary for model 2 of revised version

	Estimate	SE	t-value	p-value
Intercept	1.0396	0.2418	4.2994	0.0000
temperature	0.0170	0.0023	7.4284	0.0000
wind	-0.0521	0.0163	-3.1904	0.0019
radiation2	0.1696	0.0401	4.2324	0.0000
wind2	-0.4955	0.1637	-3.0266	0.0031
wind2×wind	0.0475	0.0184	2.5808	0.0112

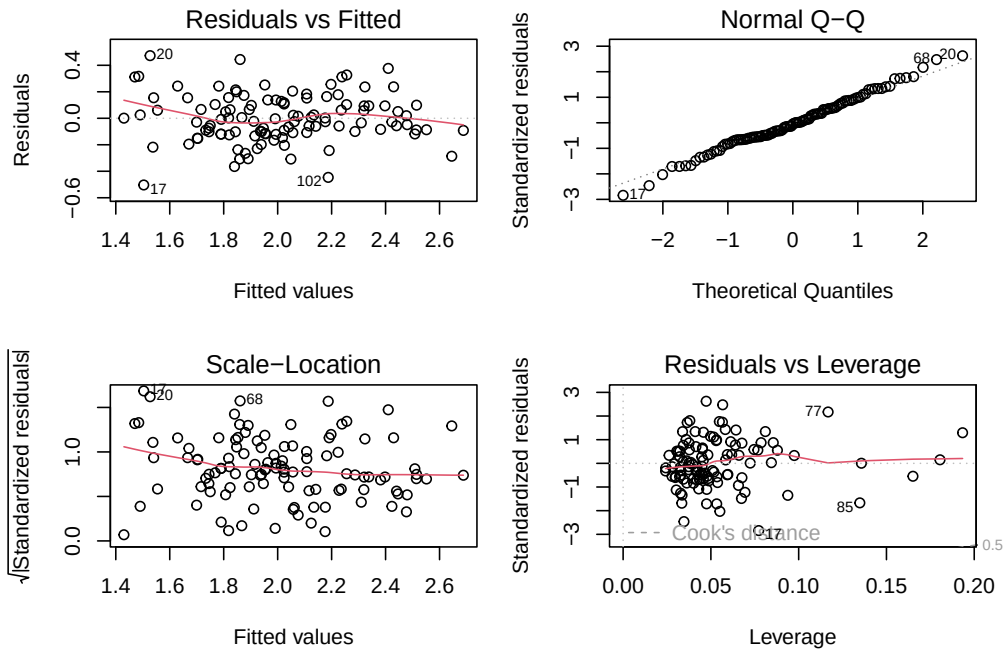


Figure 10: Residual analysis

```
resids = fit2$residuals
bp = bptest(fit2)
sh = shapiro.test(resids)
db = durbinWatsonTest(fit2)

rtest = t(round(as.data.frame(c(bp$statistic, bp$p.value, sh$statistic, sh$p.value, db$dw, db$p)),3))
rownames(rtest) = NULL

kable(
  rtest,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = rep(c("test statistic", "p-value"),3),
  align = "c",
  caption = "Model diagnostics",
  escape = FALSE
```

```
) %>%
add_header_above(c("Breusch-Pagan" = 2, "Shapiro-Wilk" = 2, "Durbin-Watson" = 2))
```

Table 9: Model diagnostics

Breusch-Pagan		Shapiro-Wilk		Durbin-Watson	
test statistic	p-value	test statistic	p-value	test statistic	p-value
8.985	0.11	0.993	0.819	1.726	0.138

```
par(mfrow = c(1,3))
for(i in 2:4){
 plot(dat1[,i], fit2$residuals, xlab = names(dat1)[i], ylab = "residuals")
}
points(dat1[dat1$wind2==1,]$wind, fit2$residuals[dat1$wind2==1], pch= 19, col = "deepskyblue3")
points(dat1[dat1$wind2==2,]$wind, fit2$residuals[dat1$wind2==2], pch= 19, col = "indianred1")
legend("topright", c("low", "high"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

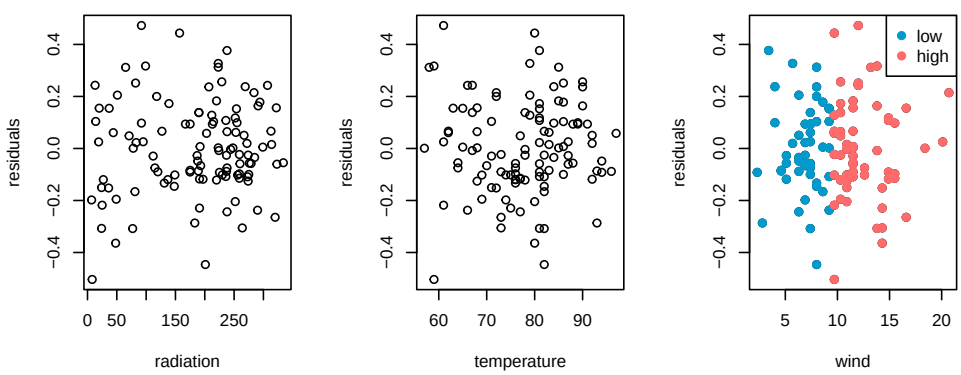


Figure 11: Residual analysis (cont.)

1.3 Model selection

The R^2, Adjusted R^2 and RMSE are used as the criterions to compare the revised Model 1 and the revised model 2. Table 10 shows the result, where one can see that the two models are comparable with each other in terms of all the criterions. However, one may think that the revised Model 2 is of higher interpretability since no any quadratic term is included in it.

```
comparison = cbind(c(summary(fit1)$r.square, summary(fit2)$r.square),
                   c(summary(fit1)$adj.r.square, summary(fit2)$adj.r.square),
                   c(sqrt(mean((predict(fit1)^5-dat1$ozone)^2)), sqrt(mean((predict(fit2)^5-dat1$ozone)
comparison = round(comparison, 4)
rownames(comparison) = c("Revised Model 1", "Revised Model 2")
```

```
kable(
  as.data.frame(comparison),
  format = "latex",
  booktabs = TRUE,
  align = "c",
  col.names = c("$\\text{R}^2$", "$\\text{Adjusted R}^2$", "RMSE"),
  caption = "Model comparison",
  escape = FALSE,
  linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 10: Model comparison

	R^2	Adjusted R^2	RMSE
Revised Model 1	0.7230	0.7098	18.5961
Revised Model 2	0.7114	0.6977	18.5773

1.4 Comments on your prediction results and scientific findings. (state at least 3 viewpoints with data evidence)

1.4.1 Comments on prediction results

The prediction results can be reflected by the RMSE shown in Table 10. One can see from Table 10 that the RMSE for both models are about 19. Comparing with the large range of ozone (see Figure 12), it is obvious that 19 can be regarded as a small number. Hence, it is reasonable to conclude that the prediction performance is quite good for both the revised Model 1 and the revised Model 2.

```
boxplot(dat1$ozone, col = "white", ylab = "ozone", horizontal = TRUE)
```

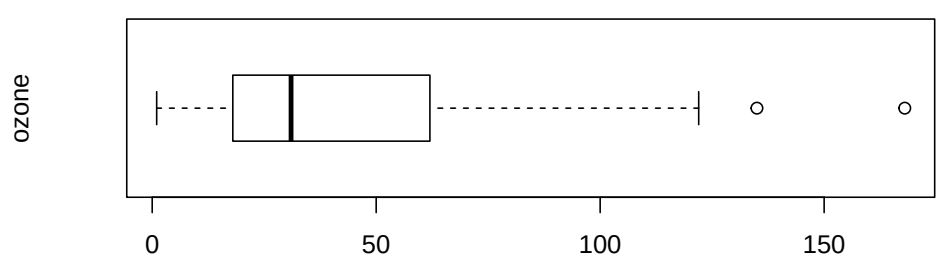


Figure 12: Boxplot of ozone

1.4.2 Scientific findings

1. According to Figure 1 and Table 8, one can infer that radiation, temperature and wind all have significant main effect on ozone. More specifically, higher radiation and higher temperature associated with higher ozone and higher wind (speed) is associated with lower ozone.

2. Table 8 shows that wind2 is significantly smaller than zero. This fact implies that perhaps the observations with lower wind (speed) and higher wind (speed) should be regarded as coming from two populations, where the ozone values observed on the former population is overall higher than those observed on the later one (reflected on the different intercepts).

3. Table 8 shows that wind2×wind is significantly greater than zero. This fact also provides evidence supporting that there may be two populations having different profiles with respect to wind (speed). More specifically, though wind has strong negative effect on ozone for the population with lower wind (speed), it has merely no any effect for those with higher wind (speed). This result is consistent with Figure 3.

---

2 Problem2

The second problem concerns about the Prostate data (more data descriptions can be found in ESL book chapter 3.2.1). There are 8 input variables and 1 response variable:

- Input variables (columns 1–8)
  - lcavol
  - lweight
  - age
  - lbph
  - svi
  - lcp
  - gleason
  - pgg45
- Response variable: lpsa (column 9)
- Indicator for training set (column 10): there are 97 observations in total, in which 70 obs'n are used as training data set and the rest of 27 observations are used as validation data. **Note: our training index is different from those used in the ESL book chapter 3.2.1.**

Goal: Predict lpsa using regression models.

Your analysis is based on the training data set only and should include the following:

(a) EDA
(b) Determine a good regression model for predicting lpsa
(c) Describe the important main effects and interaction effects.
(d) Predict lpsa for the validation data set based on the fitted model, with their prediction intervals. And compared the prediction results to the true observations. Comment on your model performance.

```r
dat2 <- read.csv("Prostate.csv")
head(round(dat2,3))
```

```
##   lcavol lweight age   lbph svi    lcp gleason pgg45   lpsa train.idx
## 1 -0.580   2.769  50 -1.386   0 -1.386       6     0 -0.431         1
## 2 -0.994   3.320  58 -1.386   0 -1.386       6     0 -0.163         0
## 3 -0.511   2.691  74 -1.386   0 -1.386       7    20 -0.163         1
## 4 -1.204   3.283  58 -1.386   0 -1.386       6     0 -0.163         1
## 5  0.751   3.432  62 -1.386   0 -1.386       6     0  0.372         0
## 6 -1.050   3.229  50 -1.386   0 -1.386       6     0  0.765         0
```

```r
dim(dat2)
```

```
## [1] 97 10
```

2.1 Exploratory Data Analysis

Data Description

This is a dataset consisting of 97 observations and 9 variables. 70 of the observations will be used as the training set, and the rest will be used as the validation set. There is no any missing value present. Two of the variables, svi and gleason, are discrete, and the rest of the variables can be regarded as continuous. To give basic understanding with regards to each variable, several descriptive statistics are shown in the following tables. For the discrete variables, Table 11 summarizes the number of observations in each level. As for the continuous ones, Table 12 summarizes the minimum, quantiles, maximum, mean and median of each of them.

```
counts = t(data.frame(c(table(dat2$svi), table(dat2$gleason))))
rownames(counts) = "counts"
kable(
  counts,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = c(names(table(dat2$svi)), names(table(dat2$gleason))),
  align = "c",
  caption = "Summary for svi and gleason",
  escape = FALSE
  ) %>%
  add_header_above(c(" " = 1, "svi" = length(names(table(dat2$svi))), "gleason" = length(names(table(da
```

Table 11: Summary for svi and gleason

	svi		gleason			
	0	1	6	7	8	9
counts	76	21	35	56	1	5

From Table 11, one can see that the sample sizes are not balanced with respect to svi and gleason. Moreover, extremely small sample sizes occur on the level 8 and level 9 of gleason. This fact suggests that the levels of gleason may need to be combined in some way so that the sample size in each level would not be too small.

```
stat = round(sapply(dat2[,-c(5,7,10)], summary),2)
kable(
  as.data.frame(stat),
  format = "latex",
  booktabs = TRUE,
  align = "c",
  col.names = colnames(dat2[,-c(5,7,10)]),
  caption = "Summary for the continuous variables",
  escape = FALSE,
  linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

From Table 12, one can see that lbpn, lcp and pgg45 seem to be somewhat right-skewed since their means are much larger than their medians in consideration of their small ranges. On the other hand, the distributions

Table 12: Summary for the continuous variables

	lcavol	lweight	age	lbph	lcp	pgg45	lpsa
Min.	-1.35	2.37	41.00	-1.39	-1.39	0.00	-0.43
1st Qu.	0.51	3.38	60.00	-1.39	-1.39	0.00	1.73
Median	1.45	3.62	65.00	0.30	-0.80	15.00	2.59
Mean	1.35	3.63	63.87	0.10	-0.18	24.38	2.48
3rd Qu.	2.13	3.88	68.00	1.56	1.18	40.00	3.06
Max.	3.82	4.78	79.00	2.33	2.90	100.00	5.58

of the variables in the rest seem to be symmetric since their means do not differ a lot form their medians in consideration of their large ranges.

Data Feature

First, the relationships between lpsa and the discrete variables are explored through visualization. Figure 13 shows the relationship between lpsa and the discrete variables. It may not be easy for one to judge whether any of the variables has effect on lpsa by inspecting on the figure. Hence numerical method is needed. Considering the small sample sizes (especially those in some of the levels of gleason) and the fact that there is no strong evidence saying that normality assumption is acceptable for lpsa under each of the levels, Wilcoxon Rank Sum test and Kruskal-Wallis test are chosen to be performed respectively for svi and gleason to examine the relationships between lpsa and them. The significance level α is set to be 0.05.

```
dat2$svi = factor(dat2$svi, levels = c(0,1))
dat2$gleason = factor(dat2$gleason, levels = c(6,7,8,9))
par(mfrow = c(1,2), mar = c(5, 4, 1.5, 2) + 0.1)
boxplot(lpsa~svi, dat2, col = "white")
boxplot(lpsa~gleason, dat2, col = "white")
```

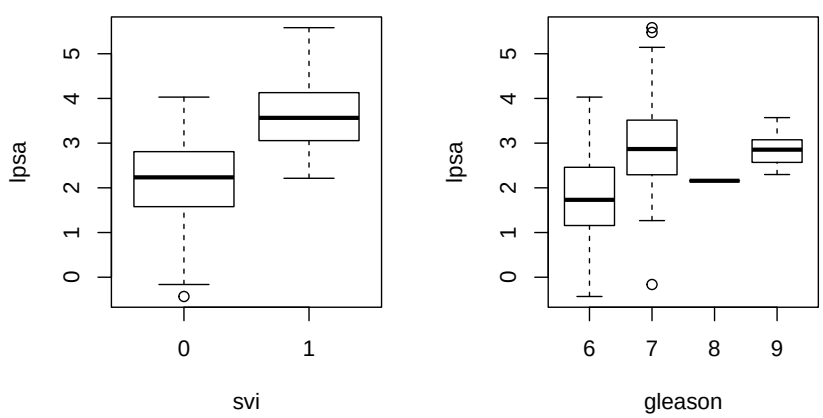


Figure 13: The relationship between lpsa and the discrete variables

Table 13 shows the values of the test statistics and p-values of the Wilcoxon Rank Sum test and the Kruskal-Wallis tests for svi and gleason. According to the p-values, one can see that svi and gleason both have effects on lpsa. Dunn's test is subsequently performed in order to further figure out how lpsa differs among the levels of gleason. The family-wise significance level α is set to be 0.05, and the p-values are computed with Bonferroni's correction.

```
w.res1 = kruskal.test(lpsa~svi, dat2)
k.res2 = kruskal.test(lpsa~gleason, dat2)
test = t(data.frame(
  round(c(w.res1$statistic, w.res1$p.value,
          k.res2$statistic, k.res2$p.value),3)))
rownames(test) = ""
kable(
  test,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = rep(c("test statistic", "p-value"),2),
  caption = "Test lpsa against the  discrete variables",
  align = "c",
  escape = FALSE
  ) %>%
  add_header_above(c(" " = 1, "svi" = 2, "gleason" = 2))
```

Table 13: Test lpsa against the discrete variables

svi		gleason	
test statistic	p-value	test statistic	p-value
30.988	0	24.666	0

```
dat2$gleason2 = factor(ifelse(dat2$gleason == 6, 0, 1), levels = c(0, 1))
```

From Table 14 and Figure 13, one can see that the lpsa for those who have higher Gleason score is significantly higher than that for those with lower Gleason score. Moreover, no significant difference of lpsa seems to be present among those with Gleason score from 7 to 9. This fact implies that it is reasonable to define a new variable named "gleason2" by renaming level 6 as "low" and combining the levels from 7 to 9 into one level named "high". Numerically, let 0 denote for the low level and 1 denote for the high level.

```
d.res2 = dunnTest(lpsa~gleason, dat2, method = "bonferroni")$res[,-c(3)]
d.res2[,2:3] = round(d.res2[,2:3], 3)
test = d.res2
kable(
  test,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = c("comparison", "test statistic", "adjusted p-value"),
  caption = "Multiple Comparison among gleason",
  align = "c",
  escape = FALSE,
  linesep = ""
  ) %>%
  kable_styling(latex_options = c("striped"))
```

Table 14: Multiple Comparison among gleason

comparison	test statistic	adjusted p-value
6 - 7	-4.839	0.000
6 - 8	-0.232	1.000
7 - 8	0.800	1.000
6 - 9	-2.336	0.117
7 - 9	-0.159	1.000
8 - 9	-0.804	1.000

Next, the basic relationships between lpsa and the continuous variables are explored through visualization and correlation analysis. According to the scatter plots in Figure 14, since many of the pairs of the variables do not seem to follow bivariate normal distribution, Kendall's τ statistic is used to measure the association among the variables. In the upper triangle of Figure 1, the values of the Kendall's τ statistics are shown. One can see that lcavol, lweight, lcp and pgg45 are significantly associated with lpsa according to Kendall's test with significance level $\alpha = 0.05$. In addition, all these associations are positive. Moreover, it is also worth noticing that some of the pairs among lcavol, lweight, lcp and pgg45 are pairs with significant correlation according to Kendall's test, implying that caution regarding to multicollinearity should be taken when fitting model.

```
pairs.panels(dat2[,-c(5,7,10:11)],
             smooth = TRUE,
             scale = FALSE,
             density = TRUE,
             ellipses = TRUE,
             method = "kendall",
             pch = 21,
             lm = FALSE,
             cor = TRUE,
             jiggle = FALSE,
             factor = 2,
             hist.col = 4,
             stars = TRUE,
             ci = TRUE,
             cex=0.8)
```

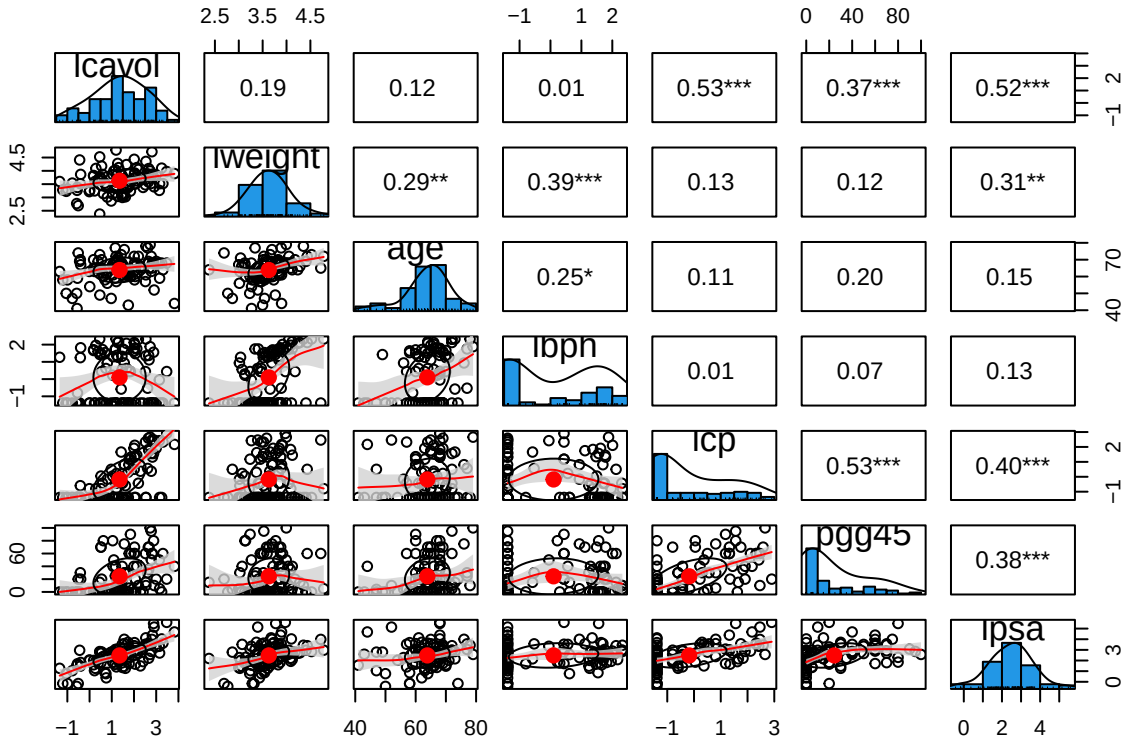


Figure 14: Correlation plot for the continuous variables

In each scatter plot in Figure 14, there is a red line showing a LOESS smoothed fitted line. When concentrating on the relationships between cavol, lweight, lcp, pgg45 and lpsa (the last row in Figure 14), one can see that the red lines are straight and upwards sloped for cavol, lweight and lcp, suggesting linear relationships between lpsa and these variables. As for pgg45, one can see that the red line is curved, suggesting curvature relationships between lpsa and pgg45. Hence, for the sake of predicting lpsa, it is reasonable to start with a linear model accounting for the relationships observed above.

In addition, from the scatter plots and the histograms, one may suspected that there may be several populations in the data which differ in the profiles with respect to lbph, lcp and pgg45. In the other words, indicator variables may be needed to discriminate out the eight populations, which can thus leave more flexibility to the model. The indicator variables named "lbph2" and "lcp2" are defined as follows: $lbph2 = 0$ if $lbph = -1.38629436$ and 1 otherwise, $lcp2 = 0$ if $lcp = -1.38629436$ and 1 otherwise. Notice that there is no need to redefine pgg45 since its distinct level occurs at 0 (see Figure 15).

```
dat2$lbph2 = factor(ifelse(dat2$lbph == -1.38629436,0,1), levels = 0:1)
dat2$lcp2 = factor(ifelse(dat2$lcp == -1.38629436,0,1), levels = 0:1)
dat2$pgg452 = factor(ifelse(dat2$pgg45 == 0,0,1), levels = 0:1)

par(mfrow =c(1,3), mar = c(5, 4, 4, 1) + 0.1)
plot(dat2$lbph, dat2$lpsa, xlab = "lbph", ylab = "lpsa", cex.lab = 1.3)
points(dat2[dat2$lbph2==0,]$lbph, dat2[dat2$lbph2==0,]$lpsa, pch= 19, col = "deepskyblue3")
points(dat2[dat2$lbph2==1,]$lbph, dat2[dat2$lbph2==1,]$lpsa, pch= 19, col = "indianred1")
legend("topright", c("0", "1"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

```
plot(dat2$lcp, dat2$lpsa, xlab = "lcp", ylab = "lpsa", cex.lab = 1.3)
points(dat2[dat2$lcp2==0,]$lcp, dat2[dat2$lcp2==0,]$lpsa, pch= 19, col = "deepskyblue3")
points(dat2[dat2$lcp2==1,]$lcp, dat2[dat2$lcp2==1,]$lpsa, pch= 19, col = "indianred1")
legend("topright", c("0", "1"), pch = 19, col = c("deepskyblue3", "indianred1"))

plot(dat2$pgg45, dat2$lpsa, xlab = "pgg45", ylab = "lpsa", cex.lab = 1.3)
points(dat2[dat2$pgg452==0,]$pgg45, dat2[dat2$pgg452==0,]$lpsa, pch= 19, col = "deepskyblue3")
points(dat2[dat2$pgg452==1,]$pgg45, dat2[dat2$pgg452==1,]$lpsa, pch= 19, col = "indianred1")
legend("topright", c("0", "1"), pch = 19, col = c("deepskyblue3", "indianred1"))
```

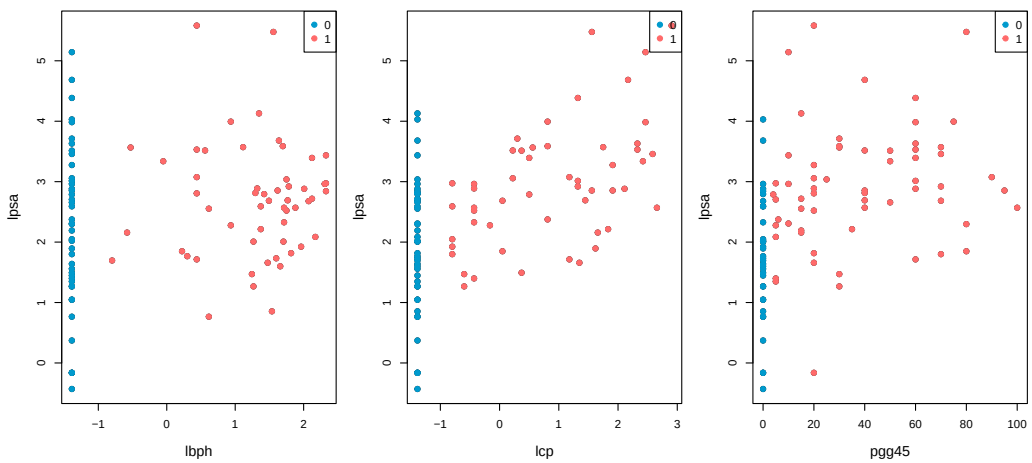


Figure 15: Conditional relationships between ozone and radiation and wind

Figure 15 shows the scatter plots in which the levels of lbph2, lcp2 and pgg45 are discriminated by different colors. For lbph and lcp, the blue points denote for the observations with level 0, and the red points denote for those with level 1. As for pgg45, the blue points denote for the observations with values equal to 0, and the red points denote for those with values greater than 0. The vertical-axis refers to lpsa in each plot. One can see that when conditioning on the level 1, though lbph still does not seem to associate with lpsa, lcp seems to be positively associated with lpsa (concentrate on the red points). In addition, pgg45 seems to be positively associated with lpsa given that its value is greater than 0. These facts imply that it is worthy to investigate a linear model in which lbph2 and lcp2 are also included.

It is natural to ask how the distributions of lbph, lcp and pgg45 would look like if the blue points shown in Figure 15 are omitted from the data. Figure 16 shows the "conditional" histograms for lbph, lcp and pgg45, where the black line refers to the density functions. One can see that there seem to be multiple modes respectively in the distributions of all the variables, implying that there may be different populations with distinct profiles in regards to lbph, lcp and pgg45. Hence, the next interest is to ask whether there is any variable in the data that is useful in discriminating and describing those latent populations.

```
par(mfrow = c(1,3))
c1 = rgb(173,216,230,max = 255, alpha = 80, names = "lt.blue")
c2 = rgb(255,192,203, max = 255, alpha = 80, names = "lt.pink")
hist(dat2[dat2$lbph2 == 1,]$lbph, freq = FALSE, col = c1, xlab = "lbph", main = "")
lines(density(dat2[dat2$lbph2 == 1,]$lbph), lwd = 2)
hist(dat2[dat2$lcp2 == 1,]$lcp, freq = FALSE, col = c1, xlab = "lcp", main = "")
lines(density(dat2[dat2$lcp2 == 1,]$lcp), lwd = 2)
```

```
hist(dat2[dat2$pgg45 > 0,]$pgg45, freq = FALSE, col = c1, xlab = "pgg45", main = "")
lines(density(dat2[dat2$pgg45 > 0,]$pgg45), lwd = 2)
```

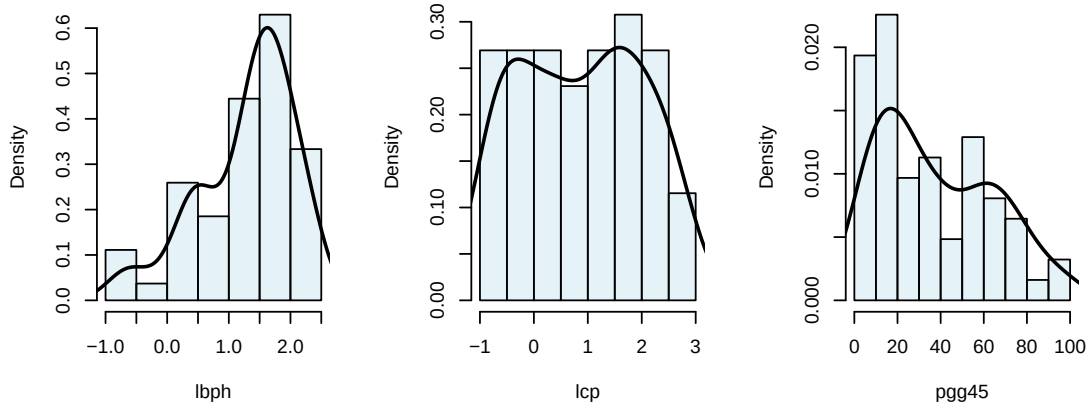


Figure 16: Conditional histograms for lcp and pgg45

```
dat2 = dat2[,-c(14)]
```

In Appendix 3.1, the conditional relationships between each variable and lbph, lcp and pgg45 are shown. It seems that the levels of lbph2, lcp2 and pgg45 can be discriminated by age. Moreover, lcavol also seems to be able to discriminate the levels of pgg45. Such facts suggest the possibility to describe and discriminate those latent populations just mentioned in terms of the conditional relationships. Combining all the findings above, one may suggest a conceptual model as follows.

$$\begin{aligned} lpsa = \beta_0 &+ \beta_1\, lcavol + \beta_2\, lweight + \beta_3\, age + \beta_4\, lbph + \beta_5\, lbph2 + \beta_6\, svi \\ &+ \beta_7\, lcp + \beta_8\, lcp2 + \beta_9\, gleason2 + \beta_{10}\, pgg45 + \beta_{11}\, pgg45^2 + \beta_{12}\, lbph2 \times lbph \\ &+ \beta_{13}\, lcp2 \times lcp + \beta_{14}\, lbph \times age + \beta_{15}\, lcp \times age \\ &+ \beta_{16}\, pgg45 \times age + \beta_{17}\, lcavol \times pgg45 + \beta_{19}\, lcp \times pgg45 \\ &+ \beta_{20}\, lcavol \times lweight + \beta_{21}\, lcavol \times age + \epsilon. \end{aligned} \tag{3}$$

Notice that the interaction terms lcavol×lweight and lcavol×age are also included to reduce the possibility of lossing any important interaction effect.

2.2 Determine a good regression model for predicting lpsa

Following the discussion above, Model (3) is fitted. Based on the fitted results, several remedial measures may be taken to revise the model. During the process, prediction performance is considered in the first place. Three assumptions are made for the error term ϵ in the models: homogeneity, normality and independence. Furthermore, the following analyses which involve hypothesis testing are all conducted under significance level $\alpha = 0.05$.

2.2.1 Model (3)

Summary for fitted Model (3)

Table 15 shows the final summary for fitted Model (3). The process was conducted by first fitting Model (3), which can be regarded as the full model. After that, sequentially remove the variable that is the least significant according to the p-values. Notice that only the one with the largest p-value would be omitted in each step. Finally, it comes up that lcavol, lweight, lcp and lcp×age are significant under significance level $\alpha = 0.05$. As for the term lcavol×lweight, though its p-value is a bit greater than 0.05, it is still decided to be kept in the model considering that the size of the training set is not very big. In the other words, it is very likely that lcavol×lweight was not significant only because that the sample size is not large enough. Svi is also kept in the model due to the similar reason. As for age, it is kept in the model since lcp×age is significant (according to the hierarchy principle).

```
fit3 = lm(lpsa~.-train.idx-gleason-lpsa-lbph2-lcp2+I(pgg45^2)+lbph2:lbph+lcp2:lcp
          +lbph:age+lcp:age+pgg45:age+lcavol:pgg45+lcavol:lweight+lcavol:age,
          data = dat2, subset = train.idx == 1)
fit3 = update(fit3, .~.-lcp2:lcp)
fit3 = update(fit3, .~.-lcavol:pgg45)
fit3 = update(fit3, .~.-I(pgg45^2))
fit3 = update(fit3, .~.-age:lbph)
fit3 = update(fit3, .~.-gleason2)
fit3 = update(fit3, .~.-lbph2:lbph)
fit3 = update(fit3, .~.-lbph)
fit3 = update(fit3, .~.-pgg45)
fit3 = update(fit3, .~.-age:pgg45)
fit3 = update(fit3, .~.-lbph)
fit3 = update(fit3, .~.-lcavol:age)
```

```
summ = as.data.frame(round(summary(fit3)$coefficients,4))
rownames(summ) = c("Intercept","lcavol","lweight", "age",
                   "svi","lcp","$\\text{lcp}\\times\\text{age}$", "$\\text{lcavol}\\times\\text{lweight
kable(summ,
  format = "latex",
  booktabs = TRUE,
  col.names = c("Estimate","SE","t-value","p-value"),
  align = c("c", "c", "c", "c"),
  escape = FALSE,
  linesep = "",
  caption = "Summary for model 3") %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 15: Summary for model 3

	Estimate	SE	t-value	p-value
Intercept	-0.7581	0.9688	-0.7825	0.4369
lcavol	1.5766	0.5806	2.7154	0.0086
lweight	0.9725	0.2521	3.8570	0.0003
age	-0.0168	0.0108	-1.5577	0.1244
svi	0.5157	0.2676	1.9274	0.0585
lcp	0.9445	0.4531	2.0847	0.0412
lcp × age	-0.0147	0.0067	-2.2075	0.0310
lcavol × lweight	-0.2774	0.1561	-1.7775	0.0804

Diagnostics for Model (3)

From both Figure 17 and Table 16, one can see that the homogeneity, normality and independence assumption all seem to be acceptable for the error term. This fact provides evidence supporting the validness of the final model. In the other words, the p-values shown in Table 15 can be regarded as correct, and meaningful inference can be made based on them.

```
par(mfrow = c(2,2), mar =c(4,4,3,1)+0.1)
plot(fit3)
```

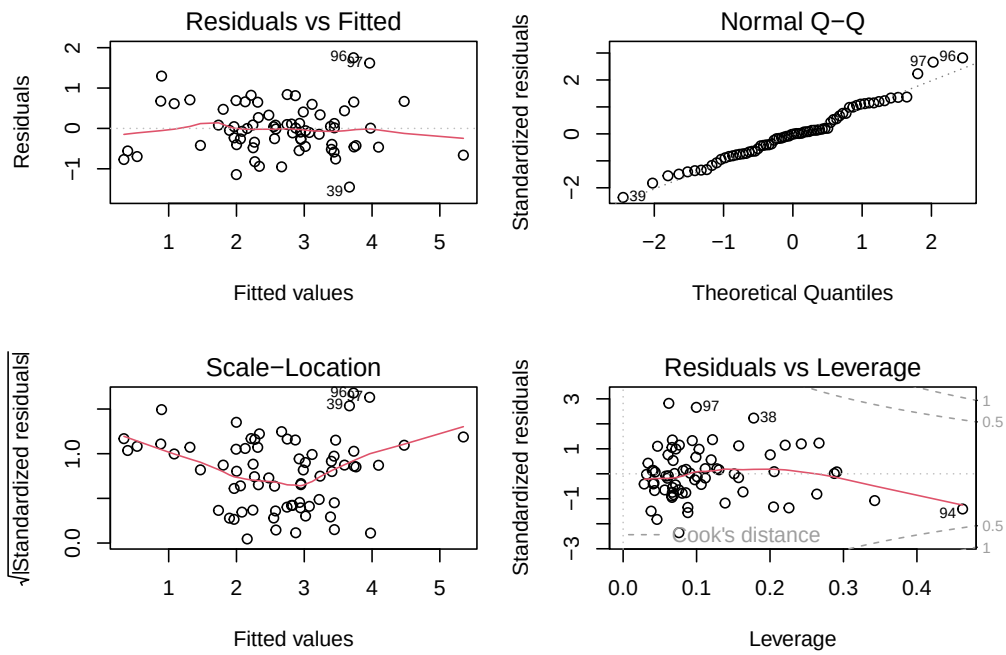


Figure 17: Residual analysis

```
resids = fit3$residuals
bp = bptest(fit3)
sh = shapiro.test(resids)
db = durbinWatsonTest(fit3)

rtest = t(round(as.data.frame(c(bp$statistic, bp$p.value, sh$statistic, sh$p.value, db$dw, db$p)),3))
rownames(rtest) = NULL

kable(
  rtest,
  format = "latex",
  longtable = TRUE,
  booktabs = TRUE,
  col.names = rep(c("test statistic", "p-value"),3),
  align = "c",
  caption = "Model diagnostics",
  escape = FALSE
) %>%
add_header_above(c("Breusch-Pagan" = 2, "Shapiro-Wilk" = 2, "Durbin-Watson" = 2))
```

Table 16: Model diagnostics

Breusch-Pagan		Shapiro-Wilk		Durbin-Watson	
test statistic	p-value	test statistic	p-value	test statistic	p-value
9.271	0.234	0.978	0.253	1.648	0.1

2.3 Describe the important main effects and interaction effects.

Main effects

According to Table 15, it is found that lcavol, lweight, age, svi and lcp have positive main effects on lpsa. The details are listed as follows in terms of the meaning of each variable.

1. Overall, those whose cancer volume is larger tends to have more prostate specific antigen.
2. Overall, those whose prostate weight is larger tends to have more prostate specific antigen.
3. Overall, those who have suffered from seminal vesicle invasion tends to have more prostate specific antigen.
4. Overall, those who have more severe capsular penetration tends to have more prostate specific antigen.

Interaction effects

According to Table 15, it is found that lcp×age and lcavol×lweight have negative interaction effects on lpsa. The details are listed as follows in terms of the meaning of each variable.

1. The same increment of capsular penetration would increase more extreme amount of prostate specific antigen on the younger patient than that on the older patient.
2. The same increment of cancer volume would increase more extreme amount of prostate specific antigen on those with lighter prostate than that on those with heavier prostate.

2.4 Predict lpsa for the validation data set based on the fitted model, with their prediction intervals. And compared the prediction results to the true observations. Comment on your model performance.

```
pred = predict(fit3, newdata = dat2[dat2$train.idx == 0, ], se.fit = TRUE, interval = "predict")
rmse = sqrt(mean((pred$fit[,1]-dat2[dat2$train.idx == 0, ]$lpsa)^2))
mape = mean(abs(pred$fit[,1]-dat2[dat2$train.idx == 0, ]$lpsa))
```

Figure 18 shows the true observations, the predicted values and the prediction interval. Moreover, the RMSE and MAPE values are also shown at the upperleft. Though the RMSE is 0.8209 and the MAPE is 0.6745, which may be regarded as small values, the width of the prediction interval seems to be a little too big in comparison with the range of lpsa. More variables may be needed to add into the model in hope of improving the prediction performance.

```
plot(dat2[dat2$train.idx == 0, ]$lpsa, ylab ="lpsa", pch = 19, col = "deepskyblue3")
points(1:27, pred$fit[,1], pch = 19, col = "indianred1")
lines(1:27, pred$fit[,2], lty = 2, col = "indianred1")
lines(1:27, pred$fit[,3], lty = 2, col = "indianred1")
text(3,4, paste("RMSE:",round(rmse,4)))
text(3,3.7, paste("MAPE:",round(mape,4)))
legend("bottomright", c("observed", "predicted", "prediction interval"),
       pch = c(19, 19, NA), lty = c(NA,NA, 2), col = c("deepskyblue3", "indianred1", "indianred1"))
```

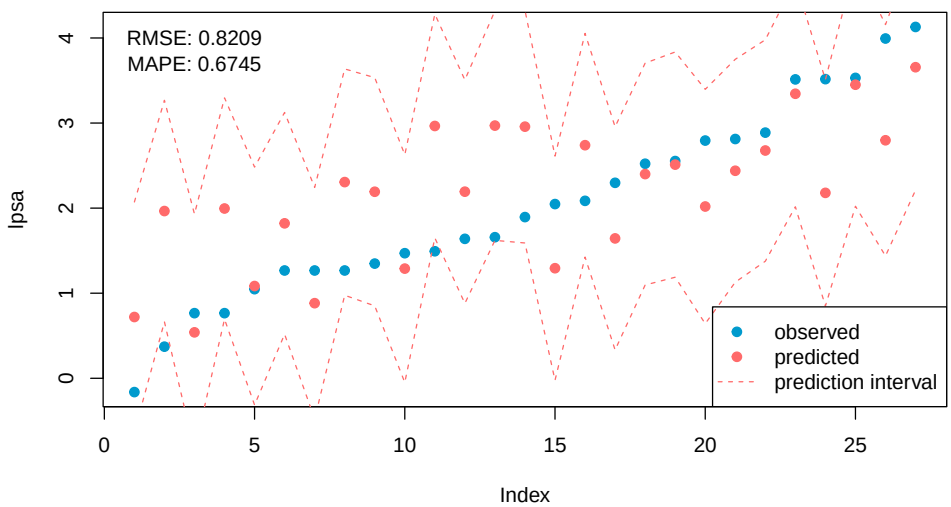


Figure 18: Prediction on the validation set

3 Appendix

3.1 The conditional relationsips between each variable and lbph, lcp and pgg45

```
temp  = dat2
par(mfrow = c(2,4))
for(i in c(1:3,5:8)){
  plot(x = temp$lbph, temp[,i], xlab = "lbph", ylab = colnames(temp)[i], pch = 19)
  points(temp[temp$lbph2 == 0,]$lbph, temp[temp$lbph2 == 0,i], col = "coral2", pch=19)
}
```

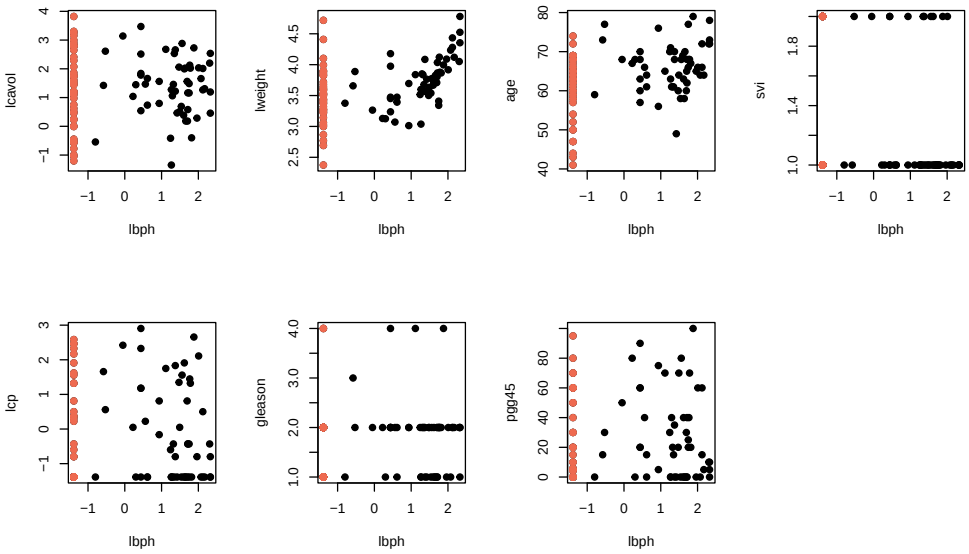


Figure 19: lbph

```
temp  = dat2
par(mfrow = c(2,4))
for(i in c(1:5,7:8)){
  plot(x = temp$lcp, temp[,i], xlab = "lcp", ylab = colnames(temp)[i], pch = 19)
  points(temp[temp$lcp2 == 0,]$lcp, temp[temp$lcp2 == 0,i], col = "coral2", pch=19)
}
```

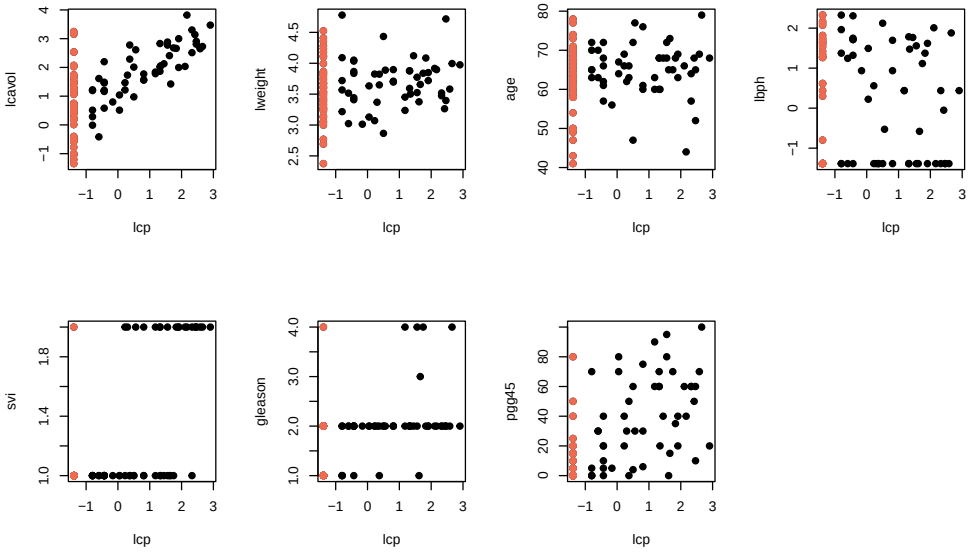


Figure 20: lcp

```r
temp  = dat2
par(mfrow = c(2,4))
for(i in c(1:7)){
  plot(x = temp$pgg45, temp[,i], xlab = "pgg45", ylab = colnames(temp)[i], pch = 19)
  points(temp[temp$pgg45 == 0,]$pgg45, temp[temp$pgg45 == 0,i], col = "coral2", pch=19)
}
```

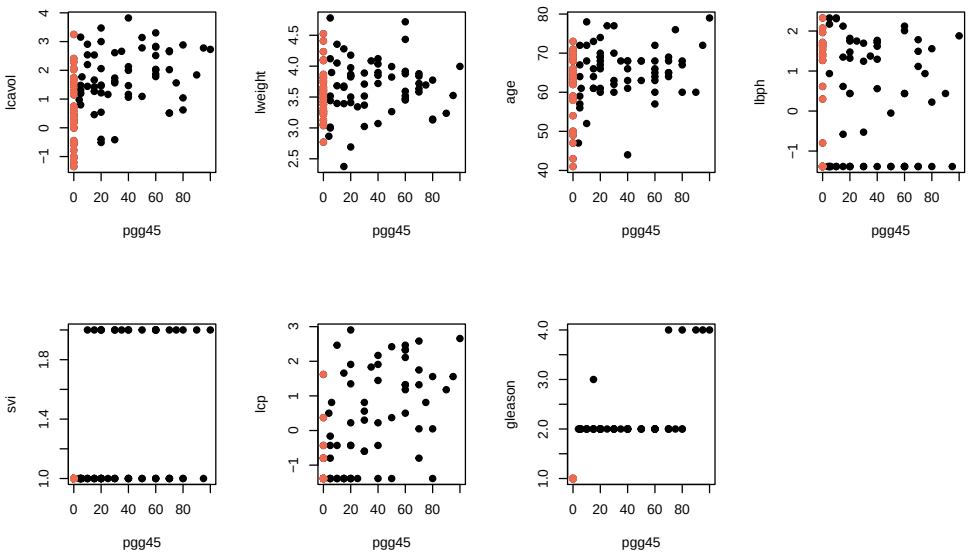


Figure 21: pgg45